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Review of One RV

EEE 554 — Lecture #2

Random Variable

A (real) random variable (RV) is a function $\underline{x} : S \rightarrow \mathbb{R}$ where S is an outcome set with probability measure P . A RV must satisfy the “measurability condition”: $\underline{x}^{-1}(A)$ must be an event in $\mathcal{F}$ for every Borel set A in $\mathbb{R}$.

Note: The measurability condition is satisfied by all RVs typically encountered in engineering.

Cumulative Distribution

The cumulative distribution function (CDF) of a RV x is defined by:

$$F_x(x) = P(\{s \in S \mid x(s) \leq x\})$$

The CDF obviously satisfies: 1. **(Monotonicity)** If $x' > x$, $F_x(x') \geq F_x(x)$ 2. **(Lower bound)** $F_x(x) \geq 0$ and $\lim_{x \rightarrow -\infty} F_x(x) = 0$ 3. **(Upper bound)** $F_x(x) \leq 1$ and $\lim_{x \rightarrow \infty} F_x(x) = 1$

Probability Density

The probability density function (PDF) of a RV x is defined by:

$$f_x(x) = \frac{d}{dx} F_x(x)$$

Existence of the PDF requires that the CDF is differentiable in some sense. Formally, the probability measure P on S must have a Radon-Nikodym derivative with respect to Lebesgue measure on $\mathbb{R}$. Typically in engineering, $\frac{d}{dx}$ is an ordinary derivative or a derivative that accommodates jump discontinuities in F_x by allowing f_x to contain Dirac delta symbols.

The PDF satisfies: 1. **(Positivity)** $f_x(x) \geq 0$ 2. **(Unit integral)** $\int_{-\infty}^{\infty} f_x(x) dx = 1$

Further, the CDF is recovered from the PDF by:

$$F_x(x) = \int_{-\infty}^x f_x(u) du$$

Probability x Takes a Value in a Set

From the definition of CDF,

$$P(\{s \mid a < x(s) \leq b\}) = F_x(b) - F_x(a) = \int_a^b f_x(x)dx$$

More generally, for a Borel set A in $\mathbb{R}$:

$$P(\{x \in A\}) = \int_A f_x(x)dx$$

Mean

The mean or expected value of x is:

$$\mu_x = E[x] = \int_{-\infty}^{\infty} x f_x(x)dx$$

Note that the mean of x may not exist.

Expectation Rule

If $y = g(x)$, then:

$$\mu_y = E[g(x)] = \int_{-\infty}^{\infty} g(x) f_x(x)dx$$

This important theorem will be proven later in this class for well-behaved transformations g .

In particular, it implies that $E[\cdot]$ is linear in the sense that:

$$\begin{aligned} E[ax + b] &= \int_{-\infty}^{\infty} (ax + b) f_x(x)dx \\ &= a \int_{-\infty}^{\infty} x f_x(x)dx + b \int_{-\infty}^{\infty} f_x(x)dx \\ &= aE[x] + b \end{aligned}$$

Variance

The variance of x is $\text{var}(x) = E[(x - \mu_x)^2]$. The expectation rule gives:

$$\begin{aligned} \text{var}(x) &= E[x^2 - 2\mu_x x + \mu_x^2] \\ &= E[x^2] - 2\mu_x E[x] + \mu_x^2 \\ &= E[x^2] - \mu_x^2 \end{aligned}$$

Moments

For $k = 1, 2, \dots$, the k^{th} moment of x is $E[x^k]$.

The k^{th} central moment of x is $E[(x - \mu_x)^k]$. Note that the first moment of x is its mean and the second central moment of x is its variance.

Characteristic Function

The characteristic function of x is:

$$\Phi_x(\omega) = E[e^{-i\omega x}] = \int_{-\infty}^{\infty} e^{-i\omega x} f_x(x) dx$$

i.e., it is the Fourier transform of the PDF.

Note that:

$$\Phi_x(0) = \int_{-\infty}^{\infty} f_x(x) dx = 1$$

and, by inverse Fourier transform:

$$f_x(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \Phi_x(\omega) e^{i\omega x} d\omega$$

Further, taking the derivative:

$$\frac{d}{d\omega} \Phi_x(\omega) = \frac{d}{d\omega} \int_{-\infty}^{\infty} e^{i\omega x} f_x(x) dx = -i \int_{-\infty}^{\infty} x e^{i\omega x} f_x(x) dx$$

Evaluating at $\omega = 0$:

$$\frac{d}{d\omega} \Phi_x(0) = -iE[x]$$

Similarly, higher moments can be generated:

$$E[x^k] = \frac{1}{(-i)^k} \frac{d^k}{d\omega^k} \Phi_x(0), \quad k = 1, 2, \dots$$